Harbour Lens Studio

Network Design Proposal | Prepared by Meverly Yong | October 2026

1. Overview

Harbour Lens Studio is a photography and design studio in Sydney with staff across two buildings. This project designs and builds a working prototype of the studio's network in Cisco Packet Tracer. Each department gets its own VLAN and subnet, staff devices receive addresses automatically, staff reach internal services by name, and guests get secure Wi-Fi that is kept away from the internal network.

2. Requirements

- Address block **10.20.0.0/24** (private range), divided with VLSM.
- Allow **10% growth** in every subnet.
- Each department on its own VLAN and subnet.
- Staff devices receive IP addresses automatically through **DHCP**.
- Staff can reach the file server by name, such as *files.harbourlens.local*, through **DNS**.
- Guest Wi-Fi is secured with **WPA2** and **cannot reach** any internal subnet (ACL).
- Hardware: one router, two switches (Building A and Building B) and one wireless access point.

3. Departments and VLANs

VLAN	Department	Devices today	Needed with 10% growth + 3 reserved	Subnet size
10	Creative (photographers and editors)	40	44 + 3 = 47	/26 (64)
20	Sales	20	22 + 3 = 25	/27 (32)
30	Admin	10	11 + 3 = 14	/28 (16)
40	Guest Wi-Fi	25	28 + 3 = 31	/27 (32)
50	Servers (DNS and file server)	5	6 + 3 = 9	/28 (16)

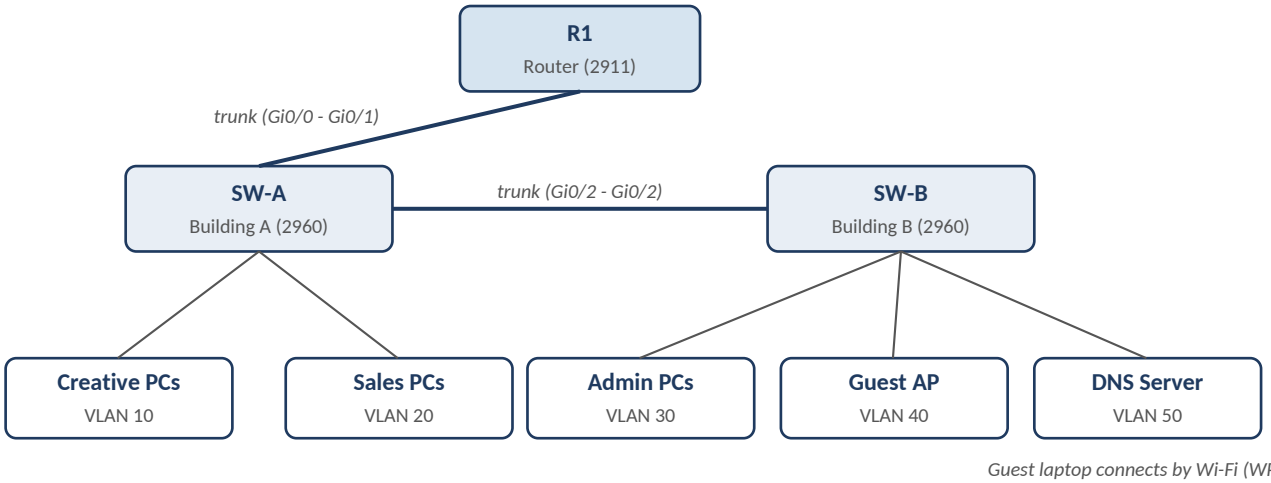
The 3 reserved addresses in each subnet are the network address, the gateway and the broadcast address.

4. IP Addressing Plan

VLAN	Department	Subnet	Mask	Gateway	Device range	Broadcast
10	Creative	10.20.0.0/26	255.255.255.192	10.20.0.1	.2 – .62	10.20.0.63
40	Guest Wi-Fi	10.20.0.64/27	255.255.255.224	10.20.0.65	.66 – .94	10.20.0.95
20	Sales	10.20.0.96/27	255.255.255.224	10.20.0.97	.98 – .126	10.20.0.127
30	Admin	10.20.0.128/28	255.255.255.240	10.20.0.129	.130 – .142	10.20.0.143
50	Servers	10.20.0.144/28	255.255.255.240	10.20.0.145	.146 – .158	10.20.0.159

Subnets are placed largest first. Addresses 10.20.0.160 – 10.20.0.255 are kept free for future departments.

5. Network Topology



A router-on-a-stick design: R1 routes between VLANs through one trunk link to SW-A, and SW-A carries all VLANs to SW-B over a second trunk.

6. Device and Port Plan

From	Port	To	Port	Mode
R1	Gi0/0	SW-A	Gi0/1	Trunk (sub-interfaces for VLANs 10 – 50)
SW-A	Gi0/2	SW-B	Gi0/2	Trunk
Creative PCs	Fa0	SW-A	Fa0/1 – Fa0/2	Access, VLAN 10
Sales PCs	Fa0	SW-A	Fa0/11 – Fa0/12	Access, VLAN 20
Admin PCs	Fa0	SW-B	Fa0/1 – Fa0/2	Access, VLAN 30
Access point	Port 0	SW-B	Fa0/11	Access, VLAN 40
DNS-Server	Fa0	SW-B	Fa0/21	Access, VLAN 50 (static 10.20.0.146)

7. Build Steps

- **Step 1 – Design:** requirements and IP plan (complete).
- **Step 2 – Topology:** place devices and cables in Packet Tracer.
- **Step 3 – VLANs and trunks:** create VLANs 10 – 50 on both switches, assign access ports, configure trunks.
- **Step 4 – Inter-VLAN routing and DHCP:** router sub-interfaces with 802.1Q tags, one DHCP pool per VLAN, gateway addresses excluded from the pools.
- **Step 5 – DNS:** static IP on the server and a DNS record for files.harbourlens.local.
- **Step 6 – Guest Wi-Fi and ACL:** WPA2 on the access point; an ACL on R1 blocks VLAN 40 from all internal subnets.
- **Step 7 – Testing:** run the test plan, take screenshots and export each device's running configuration.
- **Step 8 – Documentation:** README, diagram and lessons learned, published on GitHub.

8. Test Plan

#	Test	Expected result
1	Each PC receives an address by DHCP	Address inside its VLAN's device range
2	Ping between PCs in the same VLAN	Success
3	Ping from Creative to Sales, Admin and Servers	Success (routed through R1)
4	Ping files.harbourlens.local from a staff PC	Name resolves to 10.20.0.146 and replies
5	Guest laptop joins Wi-Fi with the WPA2 password	Receives a VLAN 40 address
6	Ping from the guest laptop to any internal subnet	Blocked by the ACL

9. Deliverables

- Packet Tracer file (.pkt) of the working network.
- Exported running configurations for R1, SW-A and SW-B.
- Topology diagram, IP plan and test screenshots.
- GitHub README explaining the design decisions, results and future improvements.